

8 (short)

- (a) List the four main microstructural mechanisms by which metals are hardened. [4]
- (b) Two cubes of aluminium of side length 1 cm are produced in annealed (soft) and cold-worked (hard) conditions. The dislocation densities in the two samples are 10^5 mm^{-2} and 10^9 mm^{-2} , respectively. Assuming a square array of parallel dislocations, estimate:
- (i) the total length of dislocation in each sample; [2]
 - (ii) the distance between dislocations in the cold-worked sample; [2]
 - (iii) the ratio of the contribution made by dislocation pinning to the yield strength of the cold-worked sample, to that of the annealed sample. [2]